

COMBINATORICS FINAL EXAM

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1. (Multinomial Coefficients) A committee of 12 people is to be divided into four subcommittees of sizes 3, 4, 2, and 3.

- (a) How many ways can this be done if the four subcommittees have distinct responsibilities?
- (b) How many ways can this be done if two of the subcommittees of size 3 have identical responsibilities, while the others are distinct?
- (c) Explain why the multinomial coefficient

$$\binom{12}{3.4.2.3}$$

appears naturally in part (a).

Solution.

□

2. (Binomial Identities) Prove the identity

$$\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{k}.$$

Give **two different proofs**

- (a) An algebraic proof using the binomial theorem.
- (b) A combinatorial proof by counting the same set in two ways.

Proof.

□

3. (Ramnsey Numbers)

- (a) Show that if $m \geq 3$ and $n \geq 3$ are integers then

$$R(m, n) \leq R(m-1, n) + R(m, n-1).$$

- (b) Show that if $R(m-1, n)$ and $R(m, n-1)$ are both even and greater than 2, then

$$R(m, n) \leq R(m-1, n) + R(m, n-1) - 1.$$

- (c) Show that $R(4, 3) = 9$, $R(5, 3) = 14$, and $R(4, 4) = 18$

Proof.

□

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4. (Stirling Numbers of the Second Kind) Let $S(n, k)$ denote the Stirling numbers of the second kind, the number of ways to partition an n -element set into k nonempty unlabeled blocks.

(a) Prove the recurrence

$$S(n, k) = kS(n-1, k) + S(n-1, k-1).$$

(b) Use the recurrence to compute $S(5, 2)$, $S(5, 3)$, and $S(6, 3)$.

(c) Give a direct combinatorial interpretation of $S(6, 3)$.

Proof.

□

5. (Pólya Enumeration) Consider coloring of the vertices of a square using three colours: red, blue, and green. Two colourings are considered equivalent if one can be obtained from the other by a rotation or reflection of the square.

(a) Identify the symmetry group acting on the vertices of the square.

(b) Write the cycle types of all symmetries of the square.

(c) Use Burnside's Lemma or Pólya's Enumeration Theorem to determine the number of distinct colourings.

(d) Briefly explain why simply computing 3^4 over counts the number of essentially different colourings.

Solution.

□

6. (Rubik's Cube Combinatorics)

(a) A standard $3 \times 3 \times 3$ Rubik's Cube has 8 corner cubies and 12 edge cubies. Ignoring the physical how many ways can the 8 corner cubies be permuted and oriented, and how many ways can the 12 edge cubies can be permuted and oriented? Find the total number of reachable $3 \times 3 \times 3$ Rubik's Cube positions.

(b) A standard $5 \times 5 \times 5$ Rubik's Cube, also called the **Professor's Cube**, as several types of movable pieces: corners, edges, wing edges, and center pieces. Find the total number of reachable $5 \times 5 \times 5$ Rubik's Cube Positions

Solution.

□

7. (Combinatorics of Continued Fractions) A finite simple continued fraction has the form

$$[a_1, a_2, \dots, a_k] = a_1 + \frac{1}{a_2 + \frac{1}{\dots + \frac{1}{a_k}}},$$

where $a_i \in \mathbb{Z}^+$.

Let $C(n)$ be the number of ordered compositions of n , that is the number of ways to write

$$n = a_1 + a_2 + \dots + a_k,$$

where $k \geq 1$ and each $a_i \geq 1$.

- Find the simple continued fraction expansion of $\frac{415}{93}$. Then verify your answer by converting the continued fraction back into a rational number.
- Find the five partial quotients in the simple continued fraction of $\sqrt{7}$. That is, write $\sqrt{7} = [a_0; a_1, a_2, a_3, a_4, \dots]$.
- Convert the periodic continued fraction $x = [2; \overline{1, 3}]$ into a quadratic irrational number in simplified radical form. That is, solve for x exactly.
- Let T_n be the number of ways to tile a $1 \times n$ board using squares of size 1×1 and dominoes of size 1×2 .
 - Prove that $T_n = T_{n-1} + T_{n-2}$, with initial values $T_0 = 1$ and $T_1 = 1$.
 - Compute T_2, T_3, T_4, T_5 , and T_6 .
 - Explain why this recurrence is related to the continued fraction

$$1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \dots}}}$$

- Show that the infinite continued fraction

$$x = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \dots}}}$$

satisfies $x = 1 + \frac{1}{x}$, and conclude $x = \frac{1+\sqrt{5}}{2}$.

Thus, the same Fibonacci-type recurrence appears in both the tiling problem and the convergents of the continued fraction.

Solution.

□

8. (Catalan Numbers) The Catalan numbers are given by

$$C_n = \frac{1}{n+1} \binom{2n}{n}.$$

- Give three different combinatorial objects counted by C_n .
- Prove the number of balanced parentheses strings with n pairs of parentheses is C_n .
- Compute C_4 , and list all balanced parenthesis strings with 4 pairs of parentheses.
- Show that the Catalan numbers satisfy the recurrence

$$C_{n+1} = \sum_{k=0}^n C_k C_{n-k}.$$

Solution.

□

9. (Generating Functions and Integer Partitions) Let $p_{\leq 4}(n)$ be the number of partitions of n into at most 4 parts.

- Write the ordinary generating function for $p_{\leq 4}(n)$.

- (b) Explain why this is also the generating function for partitions of n whose largest part is at most 4.
- (c) Use the generating function to compute $p_{\leq 4}(8)$.
- (d) List all partitions of 8 into at most 4 parts.

Solution.

□

10. (Derangements) A derangement of n objects is a permutation with no fixed points. Let D_n denote the number of derangements of n objects.

- (a) Use the inclusion-exclusion principle to prove that

$$D_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!}.$$

- (b) Compute D_5 .
- (c) Explain why D_n is the nearest integer to

$$\frac{n!}{e}.$$

Proof.

□

11. (Graph Colouring) A proper colouring of a graph is an assignment of colours to the vertices of the graph such that no two adjacent vertices (Vertices connected by an edge) share the same colour. Let C_n be the cycle graph with n vertices

- (a) Prove that the number of proper colourings of C_n using k available colours is

$$(k-1)^n + (-1)^n(k-1).$$

- (b) Use the formula to find the number of proper colourings of C_6 using 4 colours.
- (c) Explain why the answer changes depending on whether n is even or odd.

Proof.

□

12. (The Pigeonhole Principle)

- (a) Prove that among any $n+1$ integers, there must be two whose difference is divisible by n .
- (b) Suppose 51 integers are chosen from

$$[100] = \{1, 2, 3, \dots, 100\}.$$

Prove that among the chosen integers, there must be two such that one divides the other.

Proof.

□

13. (Recurrence Relations) Let a_n be the number of binary strings of length n that do not contain consecutive 1's.

- (a) Find the recurrence relation for a_n .
- (b) State the initial conditions.
- (c) Solve the recurrence or identify the sequence.
- (d) Compute a_6 and list all valid binary strings of length 4.

Solution.

□

14. (Integer Solutions and Stars and Bars) Find the number of integer solutions to

$$x_1 + x_2 + x_3 + x_4 = 20$$

under each condition.

- (a) $x_i \geq 0$ for all i .
- (b) $x_i \geq 2$ for all i .
- (c) $x_1 \geq 3$, $x_2 \geq 0$, $x_3 \geq 5$, and $x_4 \geq 1$.
- (d) $0 \leq x_i \leq 8$ for all i .

Solution.

□

15. (Set Partitions with Restricted Block Sizes) Let a_n be the number of ways to partition an n -element set into block sizes of 1, 2, or 3.

- (a) Write the exponential generating function for a_n .
- (b) Explain why the correct exponential generating function is

$$A(x) = \exp\left(x + \frac{x^2}{2!} + \frac{x^3}{3!}\right).$$

- (c) Find a summation formula for a_n .
- (d) Compute a_5 .

Solution.

□